

Selective Support-Flow Candidate Generation with Cross-Slicer Verification for Build-Orientation Search

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Abstract

Build-orientation search can reduce support material, but exhaustive slicer evaluation is expensive and cheap geometric rankings can fail on unfamiliar part families. We formulate a dimensionless Support Flow Tensor Field (SFTF v2) under normalized surface-area measure and use it selectively: an uncertainty gate proposes SFTF-ranked verification regions only when predeclared diagnostics are acceptable and otherwise abstains to a verification-budget-matched uniform search. The legacy 41-mesh collection is restricted to method development. An untouched 60-mesh Thingi10K holdout is selected by geometry-only hash ranking, and 30 of those meshes are designated in advance for an identical direction panel in CuraEngine 5.13 and PrusaSlicer 2.9.6. Corrected int32 TOMO is used as an intermediate-fidelity dense reference, not as slicer ground truth. The primary endpoint is paired grid-range-normalized regret, which remains defined for zero-support reference cases. **[LOCKED RESULT TODO: Insert the locked primary effect, confidence interval, abstention coverage, Cura and Prusa corroboration, and end-to-end runtime.]** These results determine whether SFTF is useful as a risk-controlled candidate stage rather than a replacement for broad orientation verification.

Keywords: additive manufacturing; build orientation; support material; selective prediction; abstention; CuraEngine; PrusaSlicer

1 Introduction

Build orientation controls support material, build time, surface finish, and post-processing effort in support-requiring additive manufacturing. Explicit slicer or voxel evaluations provide physically interpretable support estimates, but a dense spherical sweep is costly. Previous work has accelerated support prediction and optimized orientation by geometric, voxel, GPU, and multi-objective methods [1–4]. Our previous GPU TOMO study accelerated the verification backend [3]; the present study addresses a distinct question: which directions should be sent to that backend or to a slicer under a fixed budget?

The original SFTF formulation was promising on several organic meshes but did not outperform a budget-matched uniform search across the broader development collection. Its face-to-face coefficient also multiplied source and receiver triangle areas, producing incompatible dimensional scaling with the bed term and potential receiver-tessellation sensitivity. A router derived after viewing the same results could identify some failures but was not an external validation. These observations motivate a redesigned score and a selective policy whose abstention rule is frozen before an untouched test.

This study contributes: (i) a dimensionless SFTF defined under normalized surface-area measure; (ii) an explicit uncertainty gate with uniform fallback at an identical verification budget; (iii) a hash-locked external evaluation that retains zero-support cases; and (iv) independent corroboration using modern Cura and Prusa slicers. The central hypothesis is noninferiority to matched uniform search over all holdout meshes, with benefit assessed separately on the pre-policy accepted subset.

2 Materials and Methods

2.1 Prospective boundary and numerical provenance

All 41 meshes used while developing the legacy score, rank weights, router, or verification escalation were designated as development data. Leave-one-out and retrospective group splits within that collection were not treated as external tests. Before any new outcome was computed, 60 binary STL files were selected from a strict Thingi10K mirror by salted hash order within four face-count strata. Selection required finite geometry, positive area and bounding-box diagonal, watertightness, and one connected component. Thirty hold-out meshes were independently hash-designated for cross-slicer verification. The manifest SHA256 was e2671d0a257fd7b9437614ffc4f47df6ecfc79d15e0b7177a7335e8d2e3eb797. The complete geometry-only manifest and selection audit are reported in the Electronic Supplementary Material (ESM), which also records the sealed evaluation boundary.

Earlier TOMO grids affected by 16-bit accumulation overflow were excluded. Every end-point in this study was recomputed with a 32-bit accumulator and a recorded binary hash. **[LOCKED RESULT TODO: Insert corrected TOMO DLL hash, source commit, hardware, and reproducibility archive identifier.]**

2.2 Dimensionless surface-measure SFTF

For a triangular surface $\mathcal{S}$, one deterministic area-uniform sample $\{(\mathbf{x}_k, \mathbf{m}_k)\}_{k=1}^K$ was reused for every unit build direction $\mathbf{n}$. Let D be the bounding-box diagonal and $O_k = \max(0, -\mathbf{m}_k \cdot \mathbf{n})$, with the support-angle gate fixed during development. A ray from an active source sample in direction $-\mathbf{n}$ either intersects a valid upward-facing receiver with normal $\mathbf{r}_k$ at height h_k , or continues to the virtual build plate at height $h_{k,\text{bed}}$. The pair tensor is

$$\mathbf{F}_{\text{pair}}(\mathbf{n}) = \frac{1}{K} \sum_{k \in \mathcal{H}(\mathbf{n})} \frac{O_k}{1 + \lambda h_k / D} \mathbf{m}_k \mathbf{r}_k^T. \quad (1)$$

The scalar pair and build-plate terms are

$$R(\mathbf{n}) = \max\{0, -\mathbf{n}^T \text{sym}(\mathbf{F}_{\text{pair}}) \mathbf{n}\}, \quad (2)$$

$$B(\mathbf{n}) = \frac{1}{K} \sum_{k \in \mathcal{P}(\mathbf{n})} O_k (1 + \lambda h_{k,\text{bed}}/D), \quad (3)$$

$$S_{v2}(\mathbf{n}) = R(\mathbf{n}) + B(\mathbf{n}). \quad (4)$$

Both expectations are normalized by total sampled surface area and all lengths appear as h/D ; hence the score is dimensionless and invariant to uniform scaling. Receiver triangle area does not enter Eq. (1). The antisymmetric tensor part cannot change the Rayleigh scalar because $\mathbf{n}^T(\mathbf{F} - \mathbf{F}^T)\mathbf{n} = 0$; its norm was considered only as a predeclared uncertainty diagnostic. Python tests and an independent Mathematica script checked these identities and numerical scaling. Full derivations, deterministic implementation details, the policy-lock schema, verification-table definitions, and the reproducibility checklist are provided in the ESM.

2.3 Matched-budget selective policy

The low-score SFTF v2 basins and a deterministic near-uniform spherical design were mapped to the same corrected-TOMO orientation grid. Each branch evaluated exactly **[LOCKED RESULT TODO: insert locked number]** unique cells. The selective policy used the SFTF branch only when a fixed function of top-set sampling stability, basin crowding, score dispersion, and tensor skew ratio passed all locked thresholds; otherwise it used the uniform branch. Feature definitions, thresholds, missing-value behavior, and tie-breaking were selected using development meshes only and written to a hashed policy file before the first holdout outcome. **[LOCKED RESULT TODO: Insert the exact locked policy in an algorithm box or compact table.]**

2.4 TOMO, Cura, and Prusa verification

Corrected int32 TOMO/Tomo_Shell supplied the dense intermediate reference and the matched-budget direction values for all 60 holdout meshes. It was treated as an accelerated voxel/tomographic proxy, not as exact Cura. On the preselected 30-mesh panel, CuraEngine 5.13 and PrusaSlicer 2.9.6 both evaluated the identical union of 64 fixed Fibonacci directions, every policy-selected final direction, and fixed local offsets. Legacy CuraEngine 15.04.6 was used only as a continuity anchor.

Because slicers parameterize overhang thresholds differently, numeric angle values were not assumed equivalent. Synthetic coupons with known overhang slopes established engine-specific settings before holdout evaluation. Exact executables, profiles, configuration hashes, layer height, nozzle, material, support style, and G-code parsing rules were frozen. **[LOCKED RESULT TODO: Insert the coupon result and complete configuration table.]**

2.5 Endpoints and statistical analysis

For mesh m and policy p , the primary metric was

$$\text{NRR}_m(p) = \frac{V_m(p) - V_{m,\min}}{V_{m,\max} - V_{m,\min}}, \quad (5)$$

where the value was defined as zero when the reference range was numerically zero. Thus every holdout mesh, including zero-support references, remained in the primary population. The primary paired contrast was selective SFTF minus matched uniform NRR. Noninferiority required the upper bound of a 95% paired mesh-bootstrap interval to be below the locked margin **[LOCKED RESULT TODO: insert locked margin and engineering rationale]**. A paired permutation test and the mean, median, 90th percentile, maximum, and empirical cumulative distribution were also reported. Ratios to $V_{m,\min}$ were secondary and were computed only for strictly positive denominators. Cura and Prusa contrasts were analyzed separately; their values were not averaged into a synthetic ground truth.

3 Results

3.1 Algebraic and numerical invariance

[**LOCKED RESULT TODO:** Report symbolic identities, uniform-scaling error, remeshing sensitivity, and $K = 2048/4096/8192$ convergence. Do not proceed to performance claims if the locked tolerances fail.]

3.2 External corrected-TOMO comparison

[**LOCKED RESULT TODO:** Report all 60 meshes: paired NRR distribution, confidence interval, permutation result, maximum regret, accepted coverage, and every predeclared failure. Include uniform, SFTF-only, selective, random, and legacy ablations at matched budgets.]

3.3 Independent cross-slicer corroboration

[**LOCKED RESULT TODO:** Report Cura and Prusa separately on all 30 preselected meshes, including NRR contrasts, rank agreement, top-direction overlap, disagreement cases, and the finite-panel reference limitation.]

3.4 Computational cost

[**LOCKED RESULT TODO:** Report candidate generation, uncertainty gate, corrected-TOMO verification, Cura, and Prusa time separately, with hardware and concurrency.]

4 Discussion

The interpretation is conditional on the locked outcomes. If the primary noninferiority criterion and accepted-subset benefit are both met, the method is supported as a selective

candidate stage, not as a universal replacement for uniform verification. If only noninferiority is met, the result supports the safety of abstention but not the utility of SFTF ranking. If noninferiority fails, the correct conclusion is that the locked selective policy did not generalize. **[LOCKED RESULT TODO: Replace this decision framework with the observed outcome, without deleting adverse tails or slicer disagreement.]**

Remaining limitations include the one-hop receiver model, finite surface and orientation sampling, a geometry-only external collection without printed-part measurements, and engine-specific support-generation heuristics. The best value within a finite Cura or Prusa direction panel is not a proven global slicer optimum. The study therefore tests a reproducible candidate-and-verification pipeline under controlled settings rather than universal manufacturing quality.

5 Conclusions

[LOCKED RESULT TODO: Write one conservative paragraph after the external and cross-slicer criteria are resolved. State exactly where the policy abstained and whether it was noninferior to uniform.]

Data and code availability

[LOCKED RESULT TODO: Insert public repository DOI, immutable release tag, external manifest, policy/configuration hashes, machine-readable outcomes, and licensed data retrieval instructions. Do not redistribute restricted mesh files.]

Declaration of conflicting interests

The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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[**LOCKED RESULT TODO:** Insert the final funding statement.]

References

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